

Exercise 08 – Sensor Overview

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1 App Output

For compiling the app, I used a docker image with the android sdk and gradle installed. This allowed me to build the app without installing the android sdk on my local machine. The Google Pixel 8 Pro (Android 16). Figure 1 shows the list of all available sensors on the device.

2 Sensors for Measuring Parkinson's Disease

Parkinson's disease manifests in motor symptoms such as resting tremor, slowness of movement, rigidity and an unstable gait. These symptoms can be captured well with the motion sensors of a smartphone:

- Accelerometer
- Gyroscope
- Linear Acceleration / Gravity Sensor
- Rotation Vector / Orientation Sensor
- Step Detector / Step Counter

The **accelerometer in combination with the gyroscope** is the best choice: they directly capture the characteristic resting tremor, which can be identified with a simple frequency analysis (e.g. FFT).

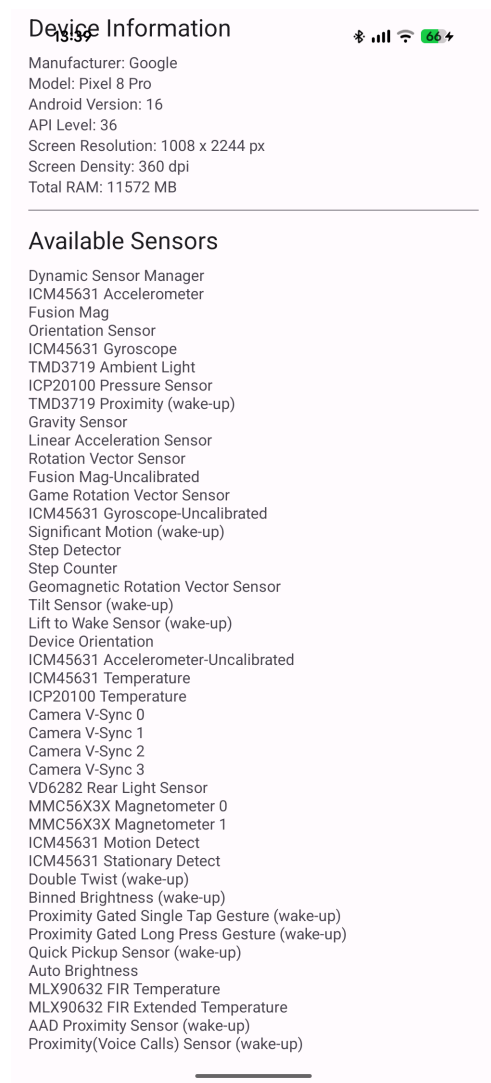


Figure 1: Output of the Sensor Overview app on a Pixel 8 Pro.